Instrumentation guide

Instrumentation (Pokhara University)

G.L. Book House

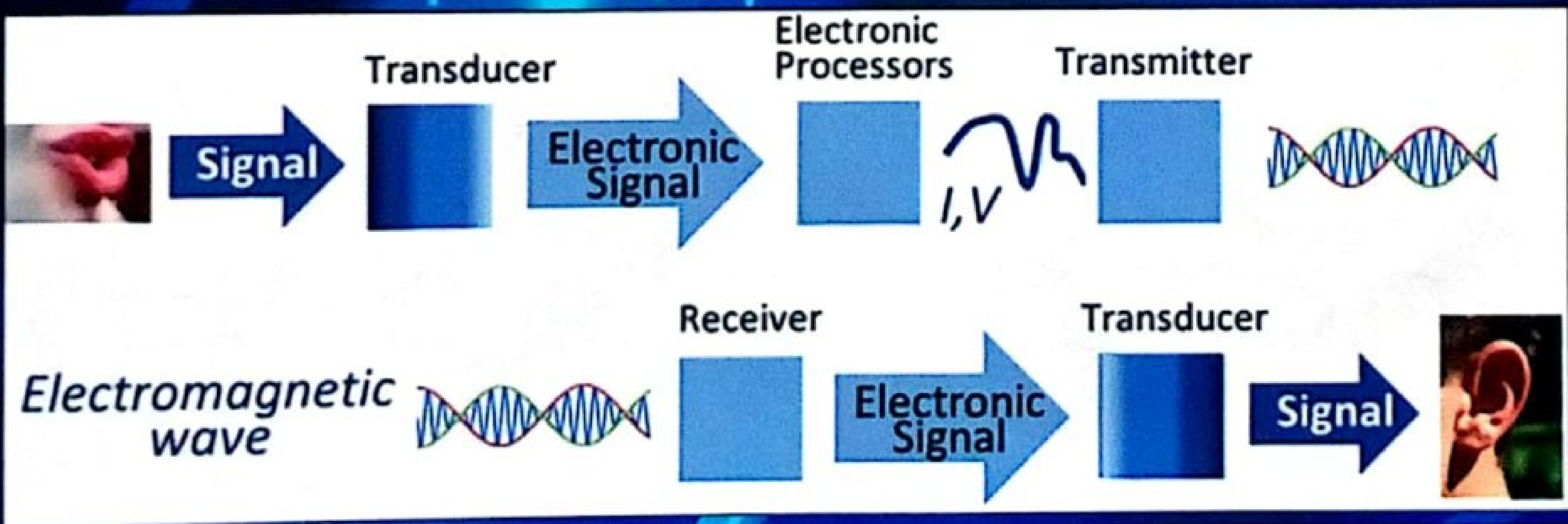
a complete manual

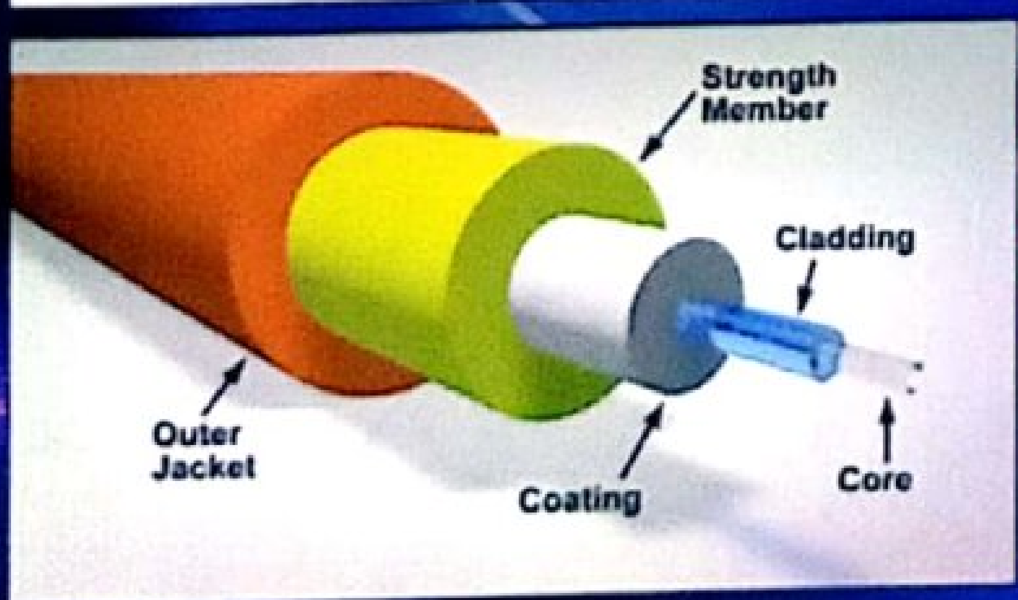
INSTRUMENTATION

Bachelor of Engineering

Pokhara & Purbanchal University

Electrical, Electronics and Communication, Computer





Er. Sanjaya Chauwal

SYLLABUS

Pokhara University Syllabus

Instrumentation (3-2-2)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Objectives:

1. To provide knowledge of instrumentation.
2. To give knowledge of measurements.
3. To develop skills of instrumentation system.

Course Contents:

1. Introduction to Instrumentation System **[3 hrs]**

- 1.1 Components of Instrumentation and their function,
- 1.2 Basic concepts of Transducer
- 1.3 Signal conditioning and transmission
- 1.4 Output device
- 1.5 Type of signals in instrumentation.

2. Signal Measurements **[12 hrs]**

- 2.1 Units and standards of measurements
- 2.2 Measuring instruments
- 2.3 Performance parameters (static and dynamic)
- 2.4 Concepts of bridges: Wheat stone bridge, Kelvin's bridge, Maxwell's bridge, Hay's bridge, Capacitance bridge, Schering bridge and Errors, Probability of errors, normal distribution

3. Physical variables and transducers **[12 hrs]**

- 3.1 Physical variables and their types (Electrical, Mechanical, process, bio-physical variable)
- 3.2 Transducer principle of operations
- 3.3 Input and output characteristics and applications of transducers (resistive, capacitive, inductive, voltage and currents)
- 3.4 Calibrations and error in transducers.

4. Signal Conditioning and processing **[8 hrs]**

- 4.1 Importance of signal conditioning
- 4.2 Signal amplification, filtering, and wave shaping
- 4.3 Instrumentation Amplifier
- 4.4 Op-Amp in instrumentation
- 4.5 Isolation amplifiers: principle and essentials of isolation amplifiers
- 4.6 Amplifier Applications

4.7 Interference signals and their elimination
4.8 Signal conversion (Analog-to-digital, Digital-to analog).

5. Data Transmission **[4 hrs]**

5.1 Transmission types
5.2 Transmission schemes
5.3 Data transmission system and standards.

6. Output Devices **[3 hrs]**

6.1 Indication instruments
6.2 Magnetic data recorders
6.3 Strip-chart, X-Y display unit and plotter.

7. Data Acquisition Systems **[3 hrs]**

7.1 Components of analog and digital Data Acquisition System
7.2 Use of Data Acquisition System
7.3 Modern trends in data acquisition system

Laboratory:

1. Conversion of physical variables into electrical signal.
2. Signal conditioning using active devices or Op-Amp.
3. Measurement of physical variables using various bridges.
4. Error measurements in instrumentation system.
5. Observation of interference in instrumentation and their remedy.
6. Transmission of signal in different mediums.
7. Conversion of analog signal into digital into analog signal.

Text Book:

A. D. Helfrick and W.D. Cooper, Modern electronic Instrumentation and Measurement Techniques, Prentice Hall of India, 1996.

Reference Books:

1. S. Wolf and R.F.M. smith, *Student Reference Manual for Electronic Instrumentation* Laboratories, Prentic Hall of India, 1996.
2. E.O. Deobelin, Measurement system: Application and Design; McGraw Hill, 1990.
3. A.K. Sawhney, *A course in Electronic Measurements and Instrumentation*, Dhanpat Rai and Sons, India, 1998.
4. C.S. Rangan, G.R. Sarma and V.S.V. Mani, *Instrumentation devices and systems*, Tata McGraw Hill, India, 1992.
5. D. M. Considine, *Process Instruments and Control handbooks*, McGraw Hill 1985.

CONTENTS

CHAPTER 1

INTRODUCTION TO INSTRUMENTATION SYSTEM

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1.1 COMPONENTS OF INSTRUMENTATION AND THEIR FUNCTION

Instrumentation is a collective term for measuring instruments used for indicating, measuring and recording physical quantities and has its origins in the art and scientific instrument-making. The main functional element of a measurement are:

i) Primary sensing element
ii) Variable conversion element
iii) Variable manipulation element
iv) Signal conditioning element
v) Data transmission element
vi) Data presentation element

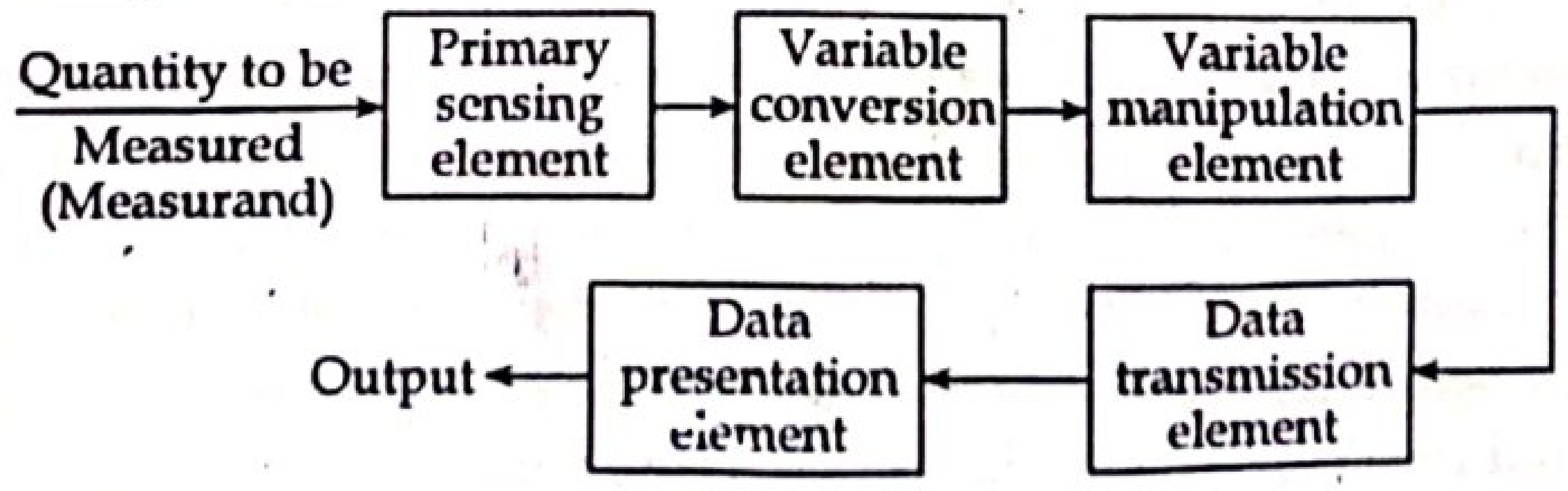

Figure: Generalized or functional block diagram of an instrumentation system.

i) Primary sensing element

The quantity or the variable which is being measured makes it first contact with the primary sensing element of a measurement system. The measurement is thus first detected by primary sensor or detector. The measurement is then immediately converted by transducer. Transducer is defined as a device which converts a physical quantity into an electrical quantity. The output of the sensor and detector element employed for measuring a quantity could be in different analogous form. This output is then converted into an electrical signal by a transducer. In many cases, the physical quantity is directly converted into an electrical quantity by a detector transducer.

For example: LDR, microphone etc

ii) Variable conversion element

The output signal of the variable sensing element may be any kind. It could be a mechanical or electrical signal. It may be a deflection of some electrical parameter such as voltage, frequency etc. Sometimes the output from the sensor is not suited to the measurement system. For the instrument to perform the desired function, it may be necessary to convert this output signal from the sensor to some other suitable form while preserving the information content of the original signal.

For example: Analog to digital conversion and vice-versa etc.

iii) Variable manipulation element

Variable manipulation means a change in numerical value of the signal. The function of a variable manipulation element is to manipulate the signal presented to this element while preserving the origin nature of the signal. The variable manipulation element could be either placed after the variable conversion element or it may precede the variable conversion element. *For example;* a voltage amplifier acts as a variable manipulation element. The amplifier accepts a small voltage signal as input and produces an output signal which is also voltage but of greater magnitude.

iv) Signal conditioning element

The output signal of transducers contain information which is further processed by the system. Many transducers develop usually a voltage or some other kind of electrical signal and quite often the signal developed is of very low voltages, may be of the order of mV and some even μV. This signal could be contaminated by unwanted signals like noise due to an extraneous source which may interface with the original output signal. Another problem is that the signal could also be distorted by processing equipment itself. If the signal after being sensed contains unwanted contamination or distortion, it is need to remove the interfering noise/ sources before its transmission to next stage otherwise we may get highly distorted results which are far from its true value.

The solution of these problems is to prevent or remove the signal contamination or distortion. The operations performed on the signal to remove the signal contamination or distortion is called signal conditioning. Many signal conditioning processes may be linear, such as amplification, attenuation, integration, differentiation, addition and subtraction. Some may be non-linear processes, such as modulation, filtering, clipping etc.

For example: filter circuit, clamper, clipper etc.

v) Data transmission element

There are several situations where the elements of an instrument are actually physically separated. In such situations it become necessary to transmit data from one element to another. The element that performs this function is called data transmission element. The signal conditioning and transmission stage is commonly known as intermediate stage.

For example: electrical cable, pneumatic pipe etc

vi) Data presentation element

The function of data presentation element is to convey the information about the quantity under measurement to the personnel handling the instrument or the system for monitoring, control or analysis purposes. The information conveyed must be in a convenient form. In case data is to be monitored, visual display devices are needed. These device may be analog or digital indicating instruments like ammeter, voltmeters etc. In case the data is to be recorded, recorders like magnetic tapes, high speed camera and TV equipment, storage type CRT, printers, analog and digital computers may be used. The final stage in a measurement system is known as terminating stage.

For example: LCD, voltmeters ammeter etc

1.2 BASIC CONCEPTS OF TRANSDUCER

Transducer is a device that converts one type of energy into other type of energy for the purpose of measurement or transfer of information. *For example;* a microphone as an input device converts sound waves into electrical signals for the amplifier to amplify and a loudspeaker as an output device converts the electrical signals back into sound waves.

Following are the factors which need to be considered while selecting a transducer;

i) High input impedance and low output impedance to avoid loading effect.

ii) Preferafly small in size.

iii) High degree of accuracy and repeatability.

iv) Selected transducer must be free from errors.

v) Good resolution over entire selected range.

vi) Highly sensitive to desired signal and insensitive to unwanted signal.

The transducer consists of two main parts:

i) **Sensing or detector element**

It is the part of the transducer which give the response to the physical sensation. The response of the sensing element depends on the physical phenomenon.

ii) **Transduction element**

The transduction element converts the output of the sensing element into an electrical signal. This element is also called the secondary transducer.

1.3 TYPES OF SIGNAL IN INSTRUMENTATION

A signal is defined as a function of one or more independent variables that contains information about the nature of some phenomenon. Voltages and currents as a function of time in an electrical circuit are examples of signals.

Classification of Signal

a) **Continuous-time and discrete-time signals**

A signal x(t) is said to be continuous-time signal if it is defined for all time t. A speech signal as a function of time is an example of a continuous-time signal.

A discrete time signal is defined only at discrete time. The weekly stock market index is an example of a discrete-time signal.

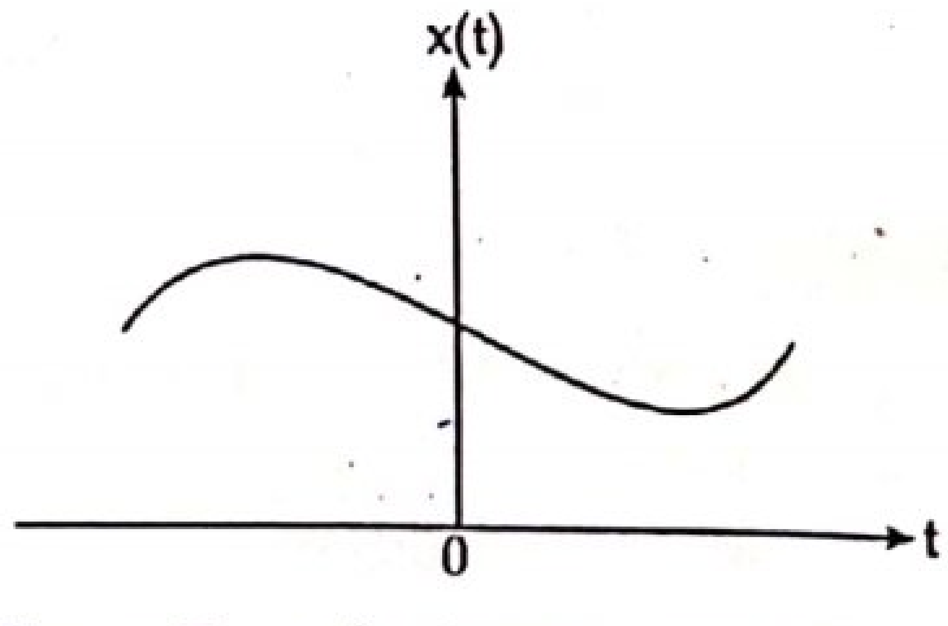

Figure: Example of continuous time signal

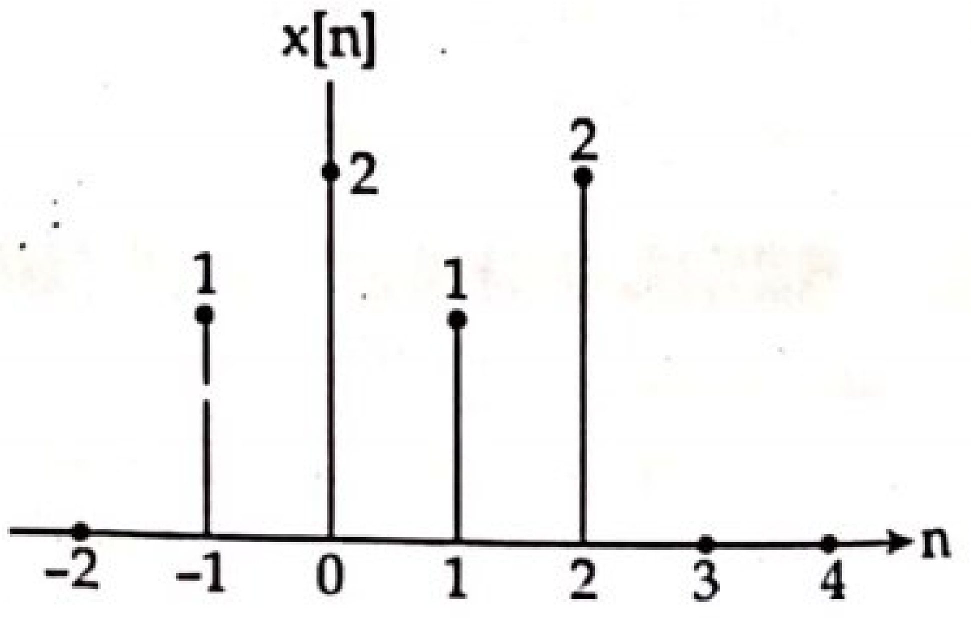

Figure: Example of discrete-time signal

Continuous time signal is denoted by x(t), where the time interval may be bounded (finite) or infinite and discrete-time signal is denoted by x[n], where n is an integer value that varies discretely.

b) **Even and odd signals**

A signal x(t) or x[n] is defined as an even signal if it is identical to its time-reversed counterpart.

i.e., $x(-t) = x(t)$

$x[-n] = x[n]$

For example: cosine wave series

A signal x(t) or x[n] is said to be an odd signal

If $x(-t) = -x(t)$

$x[-n] = -x[n]$

For example: sine wave series

Any signal can be decomposed into even and odd parts

That is,

$$x(t) = x_e(t) + x_o(t) \quad (1)$$

Substituting t by – t,

$$x(-t) = x_e(-t) + x_o(-t)$$

or, $x(-t) = x_e(t) - x_o(t)$ (2)

Solving equation (1) and (2);

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

and, $$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

Similarly for discrete time signal x[n],

$$x_e[n] = \frac{1}{2}[x[n] + x[-n]]$$

and, $$x_o[n] = \frac{1}{2}[x[n] - x[-n]]$$

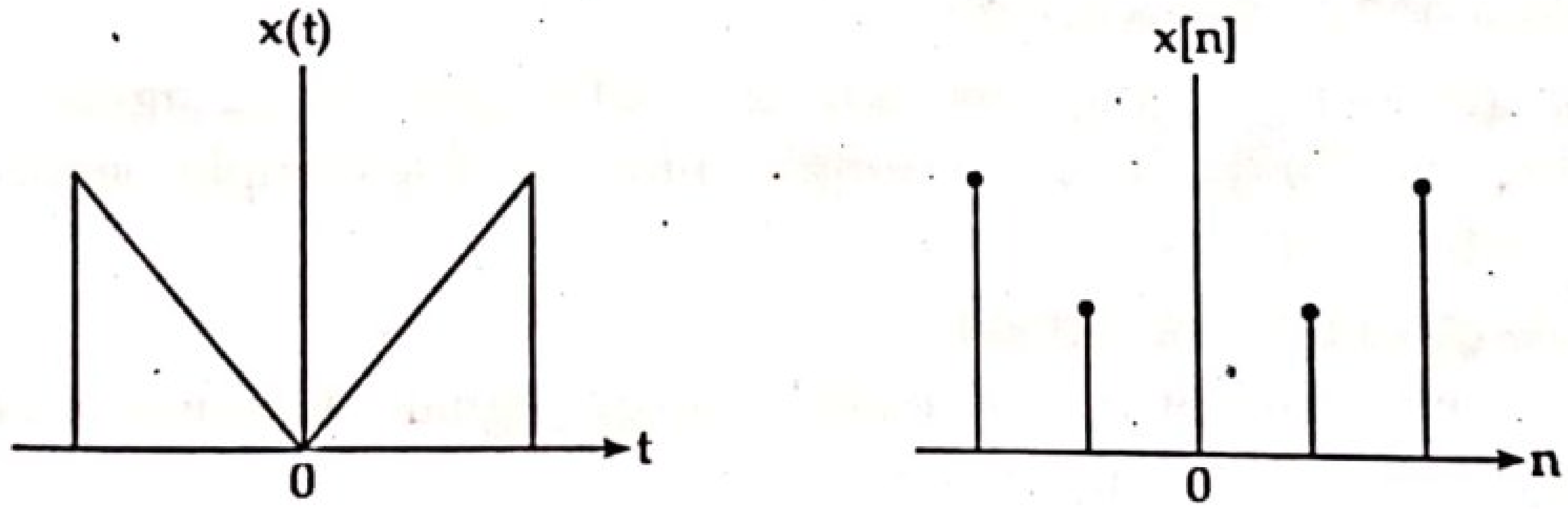

Figure: Continuous-time and discrete-time even signal.

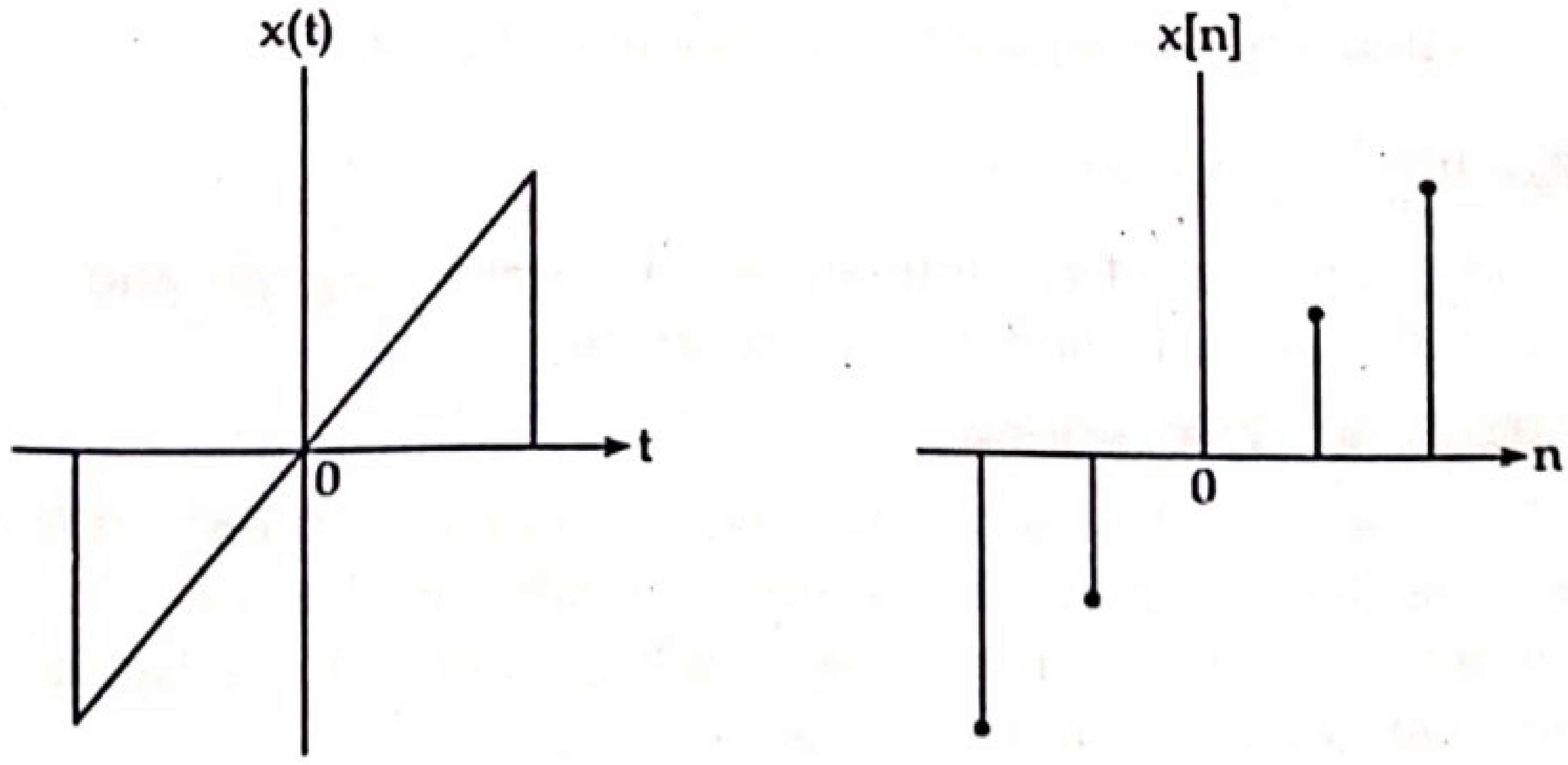

Figure: Continuous-time and discrete-time odd signal

c) Periodic and non-periodic signals

A continuous time signal x(t) is said to be periodic with period T if there is a positive non zero value of T for which $x(t + T) = x(t)$, for all values of t. Also, $x(t + mT) = x(t)$, where m is an integer.

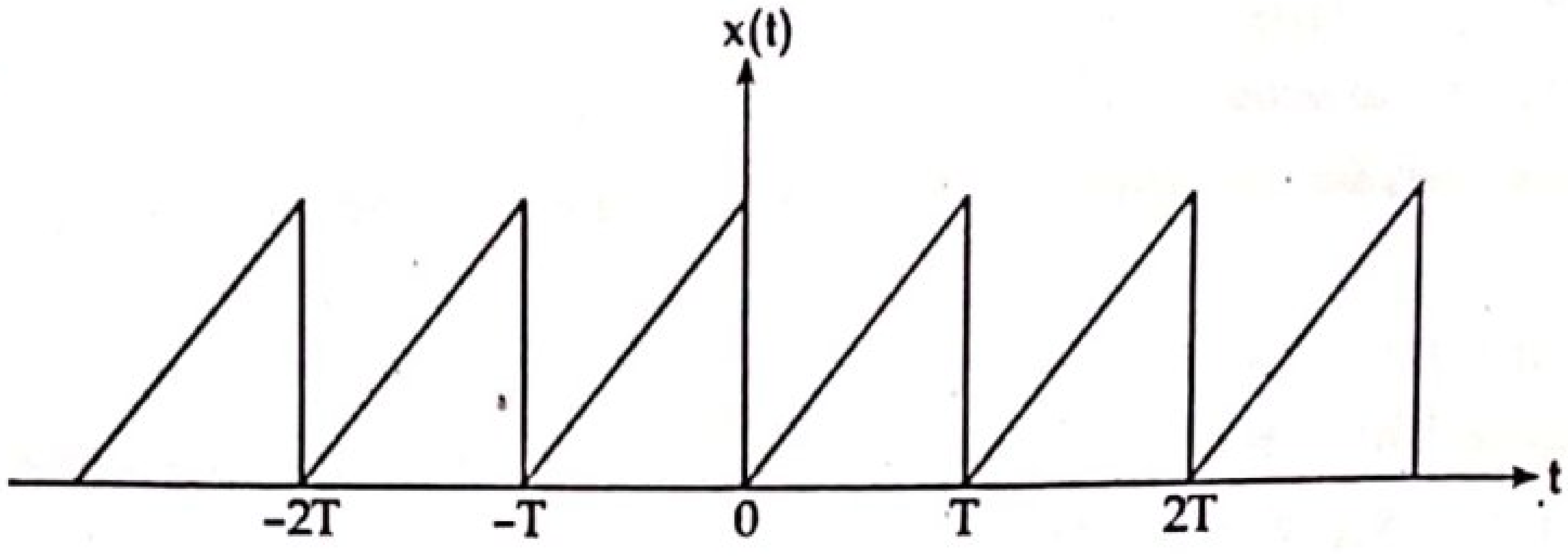

Figure: Continuous time periodic signals

The fundamental period T_0 of x(t) is the smallest positive value of T for which $x(t + T) = x(t)$ holds. A signal which is not periodic or does not hold the above condition is called the non-periodic or a periodic signal.

d) Deterministic and Random signals

The signals whose values are completely specified for any given time are deterministic signals. A deterministic signal is therefore a known function of time. *For example;* sin (ωt) is the deterministic signal.

Random signals are those signals that take random values at any given time and must be characterized statistically. Thermal noise generated in an electronic device is the random signal.

e) Real and complex signals

A signal x(t) is a real signal if its value is a real number and a signal x (t) is a complex signal if its value is a complex number. The complex signal has the form $x(t) = x_1(t) + jx_2(t)$

f) Energy and power signals

A signal with finite energy is called energy signal. A signal is called energy signal if and only if $0 < E_x < \infty$

where, $E_x = \int_{-\infty}^{\infty} x^2(t)\, dt$

A signal is called a power signal if and only if $0 < P_x < \infty$.

where, $P_x = \lim_{T\to 0} \int_{-T/2}^{T/2} x^2(t)\, dt$

Most periodic signals and random signals are power signals and most a periodic and deterministic signals are energy signals.

g) Analog and digital signals

The signal whose amplitude is continuously varying with respect to time and cannot be defined by finite number of amplitude levels are called analog signals where as the signals having only finite number of amplitude levels are called digital signals.

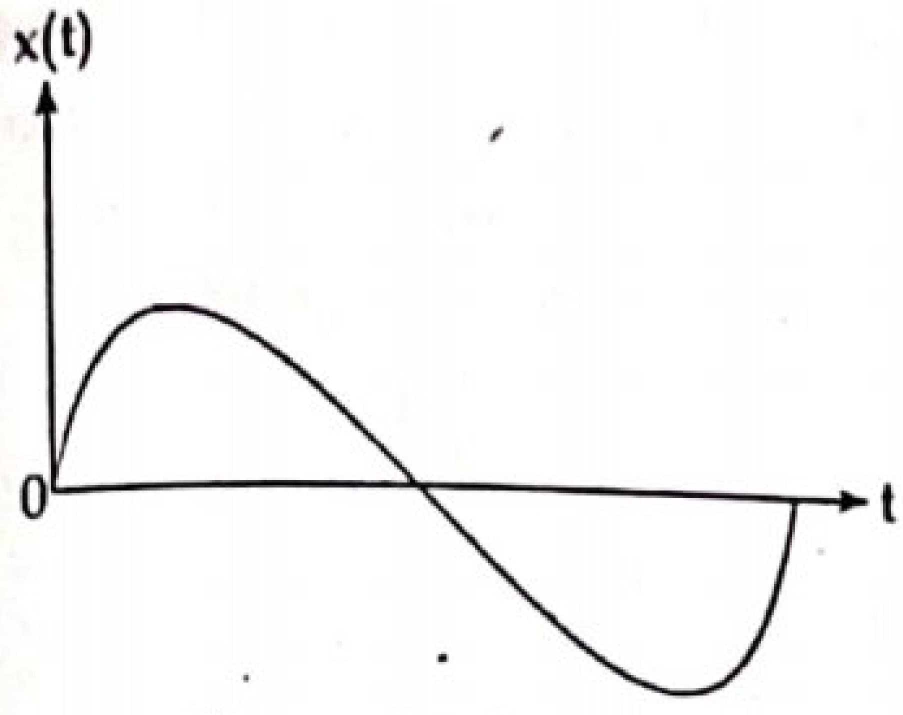

Figure: Analog signal

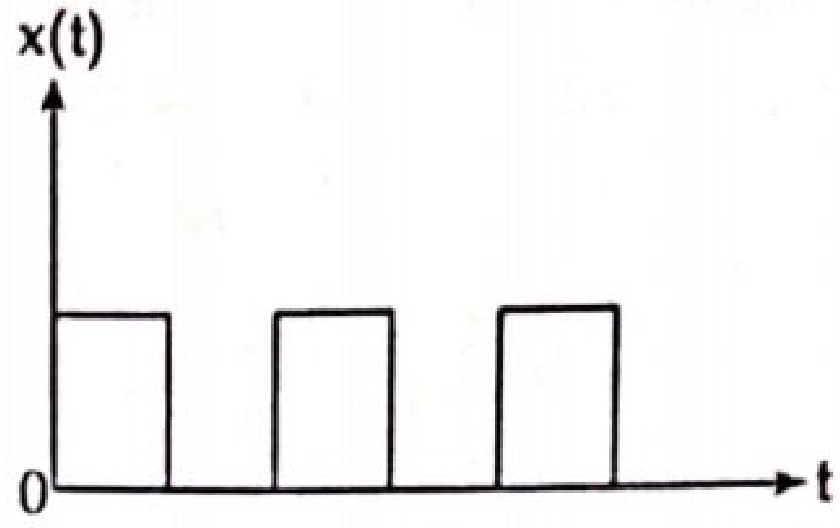

Figure: Digital signal

h) Unit step signal

A continuous time unit step signal is defined by $U(t) = \begin{cases} 1, t > 0 \\ 0, t < 0 \end{cases}$

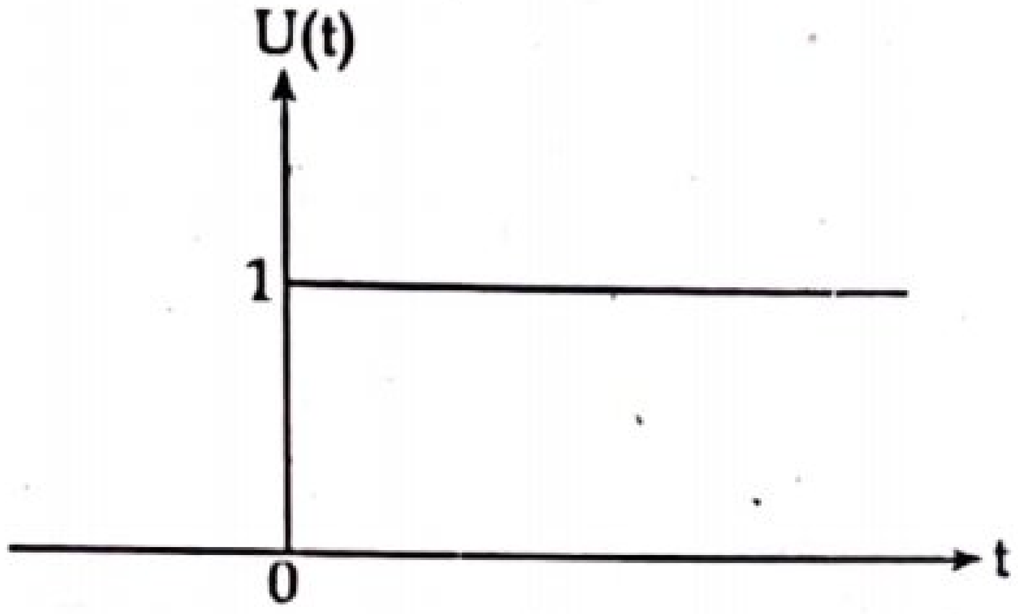

Figure: Continuous time unit step signal

i) Ramp signal

A continuous time ramp signal is defined by $r(t) = \begin{cases} t, t \geq 0 \\ 0, t < 0 \end{cases}$

Similarly, a discrete time ramp function is defined by $r[n] = \begin{cases} n, n \geq 0 \\ 0, n < 0 \end{cases}$

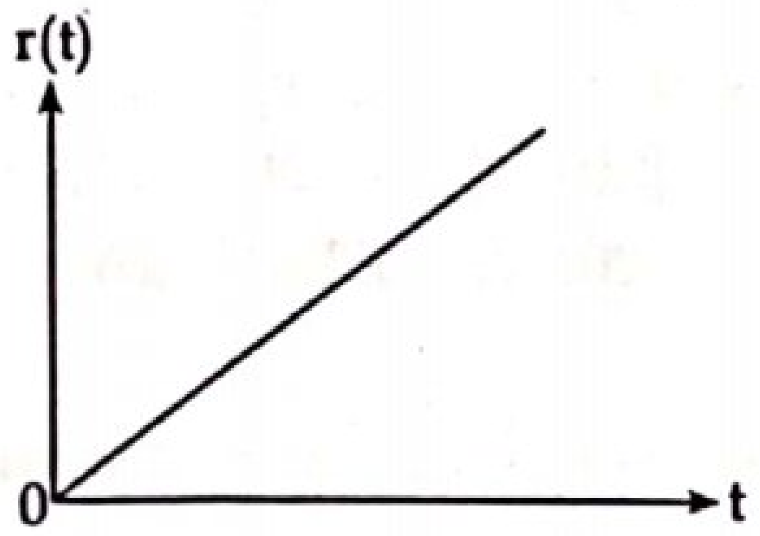

Figure: Continuous-time ramp signal

j) Unit impulse signal

A continuous time unit impulse signal is defined by $\delta(t) = 0$ for $t \neq 0$

and, $\int_{-\infty}^{\infty} \delta(t)\, dt = 1$

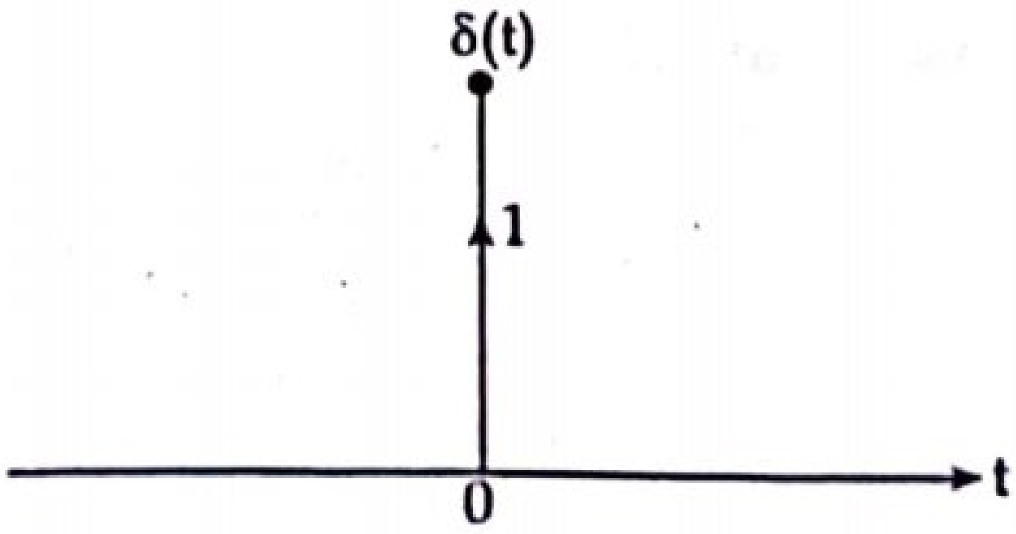

Figure: Continuous time unit impulse signal

k) **Causal, anti-causal and non-causal signals**

The signals which exist only in the positive time instants are called as causal signals. The signals that exist only in the negative time instants are called as anti-causal signals. And the signals that appear in both the positive and negative time instants are called non-causal signals.

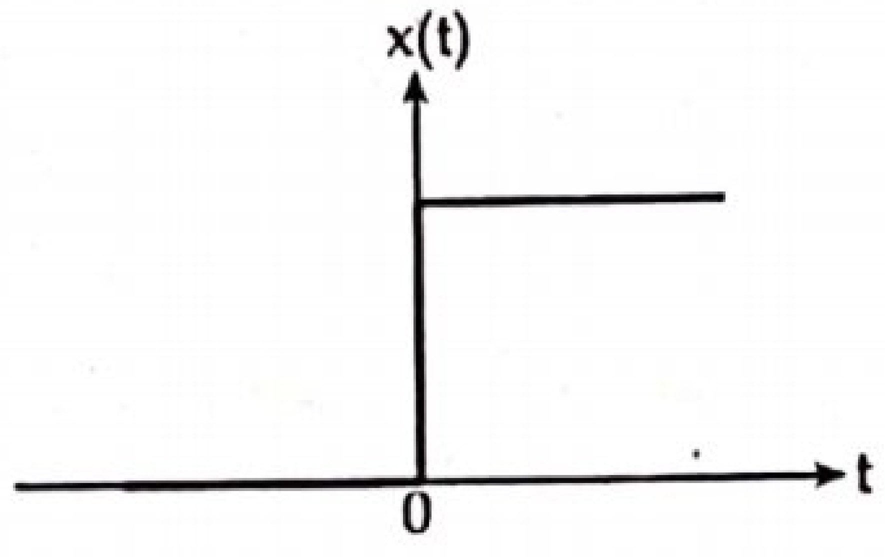

Figure: Causal signal

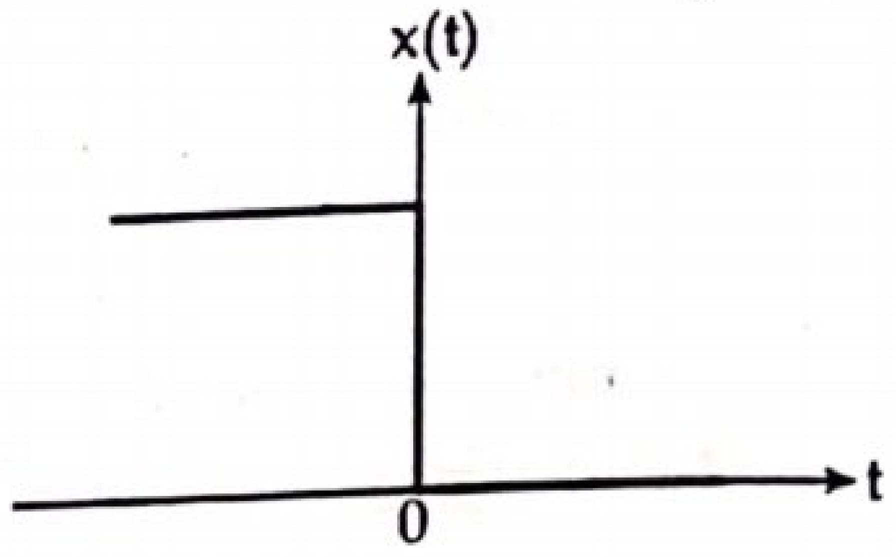

Figure: Anti-causal signal

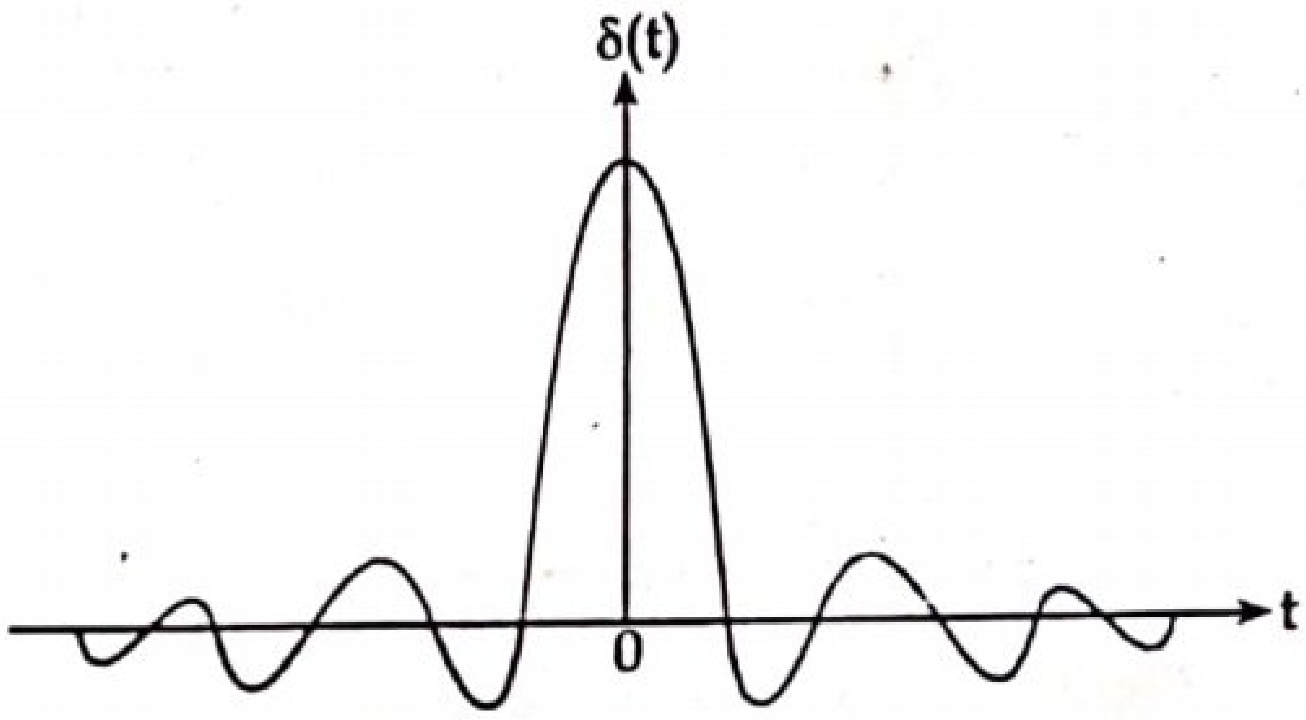

Figure: Non-causal signal

1.4 BOARD EXAM QUESTIONS SOLUTION

1. **Define instrumentation system and also explain the components of generalized instrumentation system in brief with the help of block diagram. [2011/S, 2012/F, 2012/S, 2013/S, 2014/S, 2014/F, 2015/S, 2015/F, 2017/F, 2017/S, 2018/F, 2019/F]**

Solution:

An instrumentation system is collection of instruments used to measure, monitor and control a process.

See the definition of 1.1 for components of generalized instrumentation system in brief.

2. **Define signals. Explain the different types of signals used in instrumentation system. [2011/F, 2013/F, 2016/S, 2018/S]**

Solution: See the definition of 1.3.

CHAPTER 2

SIGNAL MEASUREMENTS

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2.1 UNITS AND STANDARDS OF MEASUREMENTS

2.1.1 Units

The standard measurement of any physical quantity is known as unit. The number of times the unit occurs in any given amount of the same quantity is the number of measure. *For example*; 100 meters, we know that the meter is the unit of length and that the number of units of length is one hundred. The physical quantity, length, is therefore defined by the unit meter.

Following are the various type of units

i) Fundamental unit

The units which are independent and are not related to each other are known as fundamental unit. These units do not vary with time, temperature and pressure etc. There are seven fundamental units, as given below; Fundamental units: length, mass, time, electric current, temperature, luminous intensity and quantity of matter.

ii) Derived unit

The units which are derived from fundamental units are called derived unit. Every derived unit originates from some physical law defining that unit. This unit is recognized by its dimension, which can be defined as the complete algebraic formula for the derived unit. *For example*; the area of rectangle is proportional to its length (l) and breadth (b) or $A = l \times b$. If the meter has been chosen as the unit of length, then the unit of area is m^2. The derived unit for area (A) is then the square meter (m^2).

2.1.2 Measurement

Measurement is the act or the result of a quantitative comparison between a given quantity and a quantity of the same kind chosen as a unit. It is a process by which one can convert a physical quantity to meaningful number. It is a means of describing a natural phenomenon in quantitative term.

2.1.2.1 Standards of measurements

A standard is a physical representation of a unit of measurement. The term 'standard' is applied to a piece of equipment having a known measure of physical quantity. They are used for the purpose of obtaining the values of the physical properties of other equipment by comparison methods. In fact, a unit is realized by reference to a material standard or to a natural phenomenon including physical and atomic constants. *For example*; the fundamental unit of mass in the metric system (SI) is the kilogram, 'defined as the mass of the cubic decimeter of water at its temperature of maximum density of 4°C.

Types of standards of measurements are

i) International standards
ii) Primary standards
iii) Secondary standards
iv) Working standards

I) International standards

The international standards are defined on the basis of international agreement. They represent the units of measurements which are closest to the possible accuracy attainable with present day technological and scientific methods. International standards are checked and evalvated regularly against absolute measurements in terms of the fundamental units. The international standards are maintained at the international bereave of weights and measures and are not available to the ordinary user of measuring instruments for the purpose of calibration or comparison. Improvements in the accuracy of absolute measurements have made the international units superfluous and they have been replaced by absolute units. One of the main reason for adopting an absolute system of units is that now wire resistance standards can be constructed which are sufficiently permanent and do not vary appreciably with time.

ii) Primary standards

Primary standards are absolute standards of such high accuracy that they can be used as the ultimate reference standards. These standards are maintained by national standards laboratories in different parts of the world. The primary standards which represent the fundamental units and some of the derived electrical and mechanical units are independently calibrated by absolute measurements at each of the national laboratories. The results of these measurements are compared against each other, leading to a world average figure for the primary standards. Primary standards are not available for use outside the national laboratories. One of the main functions of the primary standards is the verification and calibration of secondary standards.

The primary standards are few in number. They must have the highest possible accuracy. Also these standards must have the highest stability *i.e.*, their values should vary as small as possible over long periods of time even if there are environmental and other changes.

iii) Secondary standards

The secondary standards are the basic reference standards used in industrial measurement laboratories. The responsibility of maintenance and calibration of these standards lies with the particular industry involved. These standards are checked locally against reference standards available available in the area secondary standards are normally sent periodically to the national standards laboratories for calibration and comparison against primary standards. The secondary standards are sent back to the industry by the national laboratories with a certification as regards their measured values in terms of primary standards.

iv) Working standards

The working standards are the major tools of a measurement laboratory. These standards are used to check and calibrate general laboratory instruments for their accuracy and performance. *For example;* a manufacturer of precision resistances, may use a standard resistance (which may be a working standard) in the quality control department for checking the values of resistors that are being manufactured. This way, he verifies that his measurement set up performs within the limits of accuracy that are specified.

v) IEEE standards

This standard is maintained by the institute of electrical and electronics engineers, IEEE an engineering society headquartere in New York city. These standards are not physical items that are available for comparison and checking of secondary standards but are standard procedures, nomenclature, definitions etc. It gives the standard test method for testing and evaluating various electronic systems and components. *For example;* there is a standard method for testing and evaluating attenuators.

2.2 MEASURING INSTRUMENTS

It is a device for determining the magnitude of a physical quantity being measured. All measuring instruments are subject to varying degrees of instrument error and measurement uncertainty. There are two main types of the measuring instruments; analog and digital.

The analog instruments indicate the magnitude of the quantity in the form of the pointer movement. The digital measuring instruments indicate the values of the quantity in digital format that is in numbers which can be read easily.

Measuring instruments may be divided into two categories, *i.e.,*

a) Absolute instrument
b) Secondary instrument

a) Absolute instrument

It gives the quantity to be measured in terms of instrument constant and its deflection.

b) Secondary instrument

In secondary instrument, the deflection gives the magnitude of electrical quantity to be measured directly. These instruments are required to be calibrated by comparing with another standard instrument before putting into use secondary instruments can be classified into three types.

i) Indicating instruments

It indicates the magnitude of an electrical quantity at the time when it is being measured. The indicators are given by a pointer moving over a graduated dial.

For example; ammeters, watt meters etc.

ii) Recording instruments

The instruments which keep a continuous record of the variations of the magnitude of an electrical quantity to be observed over a defined period of time.

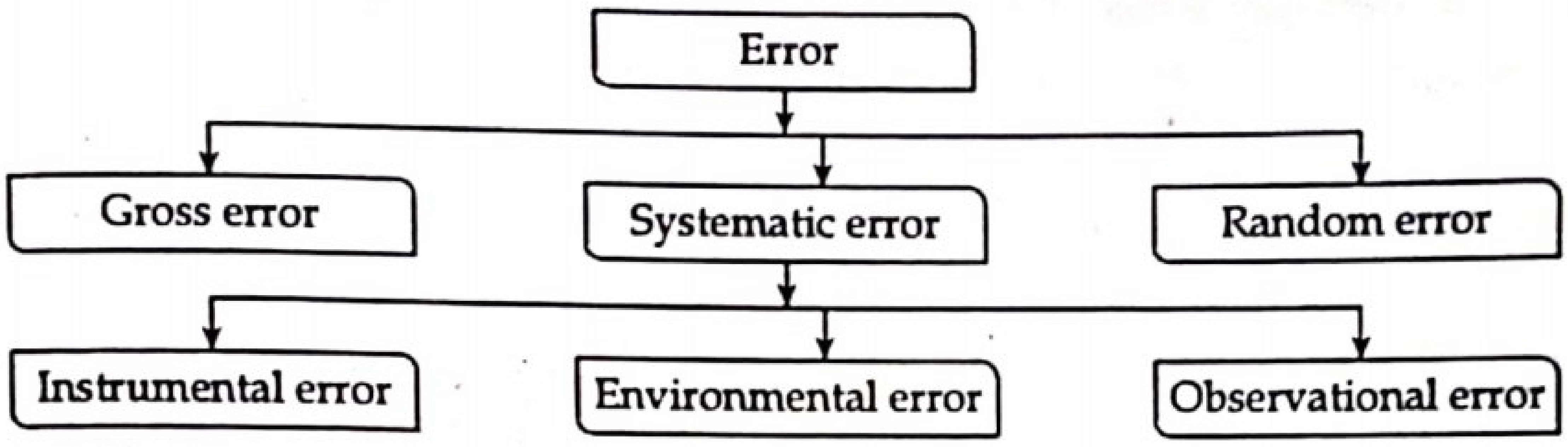
iii) Integrating instruments

The instruments which measure the total amount of either quantity of electricity or electrical energy supplied over a period of time.

For example; ampere-hour meter, energy meter etc.

2.2.1 Classification of Error

No measurement can be made with perfect accuracy but it is important to find out what accuracy actually is and how different errors have entered into the measurement. A study of errors is a first step in finding ways to reduce them. Errors may arise from different sources and are usually classified as under.



a) Gross error

This class of errors mainly covers human mistakes in reading instruments and recording and calculating measurement results. This responsibility of the mistake normally lies with the experimenter. The experimenter may grossly misread the scales. As long as human beings are involved, some gross wilt definitely be committed. Although complete elimination of gross errors is probably impossible, one should try to anticipate and correct them. Some gross errors are easily detected while others may be very difficult to defect.

Gross errors may be of any amount and therefore their mathematical analysis is impossible. However, they can be avoided by adopting two means. They are:

i) Great care should be taken in reading and recording the data.

ii) Two, three or even more readings should be taken for the quantity under measurement. These readings should be taken preferably by different experimenters and the readings should be taken at a different reading point to avoid re-reading with the same error. It should be understood that no reliance be placed on a single reading. It is always advisable to take a large number of readings as a close agreement between readings assures that no gross error has been committed.

b) Random (Residual) errors

It has been consistently found that experimental results show variation from one reading to another, even after all systematic errors have been accounted for. These errors are due to a multitude of small factors which change or fluctuate from one measurement to another and are due surely to change. The quantity being measured is affected by many happenings throughout the universe. We are aware and account for some of the factors influencing the measurement, but about the rest we are unaware. The happenings or disturbance about which we are unaware are lumped together are called 'Random' or 'Residual.' Hence the errors caused by these happenings are called random errors.

Random errors can be eliminated by increasing the number of readings and using statistical means to obtain the best approximate of true value of the quantity under measurement.

c) Systematic errors

These types of errors are divided into three categories:

i) **Instrumental Errors**

These errors arise due to three main reasons:

- Due to missue of the instruments
- Due to loading effect of instruments
- Due to inherent short comings in the instrument

Elimination Method:

- The procedure of measurement must be carefully planned. Substitution methods or calibration against standards may be used for the purpose.
- The instrument may be re-calibrated carefully.
- Correction factors should be applied after determining the instrumental errors.
- Errors caused by loading effects of the meters can be avoided by using them intelligently.

ii) **Observational Errors**

There are many sources of observational errors. There are human factors involved in measurement. The sensing capabilities of individual observers effect the accuracy of measurement. No two persons observe the same situation in exactly the same way where small details are concerned. Modern electrical instruments have digital display of output which completely eliminates the errors on account of human observational or sensing powers as the output is in the form of digits.

iii) **Environmental Errors**

These errors are due to conditions external to the measuring device including conditions in the area surrounding the instrument. These may be effects of temperature, pressure, humidity, dust, vibration or of external magnetic or electrostatic fields.

Elimination Method:

- Arrangements are made to keep the conditions as nearly as constant as possible. *For example;* temperature can be kept constant by keeping the equipment in a temperature controlled enclosure.
- Using equipment which is immune to these effects. *For example;* variation in resistance with temperature can be minimized by using resistance materials which have a very low resistance temperature coefficient.

- Employing techniques which eliminate the effects of these disturbances. *For example*; the effect of humidity dust etc can be entirely eliminated by hermetically sealing the equipment.
- Applying computed corrections: Efforts are normally made to avoid the use of application of computed corrections, but where these corrections are needed and are necessary, they are incorporated for the computations of the results.

2.2.2 Statistical Analysis

a) Arithmetic mean

The arithmetic mean is given by;

$$\bar{x} = \frac{(x_1 + x_2 + x_3 + x_4 + \dots\dots\dots + x_n)}{n} = \frac{\Sigma x}{n} = \frac{\text{Sum of values}}{\text{Count of values}}$$

where, $\bar{x}$ = arithmetic mean

n = number of readings

$x_1, x_2, \dots\dots, x_n$ = Readings taken

b) Range

The simplest possible measure of dispersion is the range which is the difference between greatest and least values of data.

c) Deviation

Deviation is departure of the observed reading from the arithmetic mean of the group of readings. Let the deviation of reading x_1 be d_1 and that of reading x_2 be d_2, etc, then

$$d_1 = x_1 - \bar{x}$$

$$d_2 = x_2 - \bar{x}$$

$$\dots = \dots\dots$$

$$d_n = x_n - \bar{x}$$

and, $\bar{x} = \dfrac{\Sigma(x_n - d_n)}{n}$

The algebraic sum of deviation is zero.

where, d_n is the deviation of the n^{th} reading from the mean.

d) Average deviation

The average deviation is an indication of the precision of the instruments used in making the measurements. Highly precise instruments yields a low average deviation between readings. It is defined as the sum of the absolute values of deviations divided by the number of readings. It may be expressed as,

$$D = \frac{|d_1| + |d_2| + \dots\dots + |d_n|}{n} = \frac{\Sigma|d|}{n}$$

e) Standard Deviation (SD)

The standard deviation of an infinite number of data is defined as the square root of the sum of the individual deviations squared, divided by the number of readings.

$$SD = \sqrt{\frac{d_1^2 + d_2^2 + \dots\dots\dots + d_n^2}{n}} = \sqrt{\frac{\Sigma d^2}{n}}$$

In practice, however, the number of observations is finite. When the number of observations is greater than 20, SD is denoted by symbol σ while if it is less than 20, the symbol used is S. The standard deviation of finite number of data is given by

$$S = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + \dots\dots\dots d_n^2}{n - 1}}$$

$$\sigma = \sqrt{\frac{\Sigma d^2}{n - 1}}$$

f) Variance or mean square deviation

The variance is the mean square deviation which is the same as SD except that square root is not extracted.

Variance, V = (Standard deviation)^2

$$= (SD)^2 = \sigma^2 = \frac{d_1^2 + d_2^2 + d_3^2 + \dots\dots\dots + d_n^2}{n} = \frac{\Sigma d^2}{n}$$

But when the number of observations is less than 20, variance,

$$V = S^2 = \Sigma \frac{d^2}{n - 1}$$

Example 1

A set of independent current measurements were taken by six observers and were recorded as 12.8 A, 12.2 A, 12.5 A, 13.1 A, 12.9 A and 12.4 A. Calculate,

i) Arithmetic mean
ii) Deviation from the mean
iii) Standard deviation
iv) Variance
v) Average deviation:

Solution:

i) Arithmetic mean,

$$\bar{x} = \frac{\Sigma x}{n} = \frac{12.8 + 12.5 + 13.1 + 12.9 + 12.4}{6}$$

$$\bar{x} = 12.65 \text{ A}$$

ii) Deviations are,

$$d_1 = x_1 - \bar{x} = 12.8 - 12.65 = 0.15 \text{ A}$$

$$d_2 = x_2 - \bar{x} = 12.2 - 12.65 = -0.45 \text{ A}$$

$$d_3 = x_3 - \bar{x} = 12.5 - 12.65 = -0.15 \text{ A}$$

$$d_4 = x_4 - \bar{x} = 13.1 - 12,65 = 0.45 \text{ A}$$

$$d_5 = x_5 - \bar{x} = 12.9 - 12.65 = 0.25\ A$$

$$d_6 = x_6 - \bar{x} = 12.4 - 12.65 = -0.25\ A$$

iii) Average deviation,

$$D = \frac{\Sigma|d|}{n} = \frac{0.15 + 0.45 + 0.15 + 0.45 + 0.25 + 0.25}{6} = 0.283$$

For average deviation, absolute value is taken.

iv) Standard deviation,

$$S = \sqrt{\frac{\Sigma d^2}{n-1}}$$

$$= \sqrt{\frac{(0.15)^2 + (-0.45)^2 + (-0.15)^2 + (0.45)^2 + (0.25)^2 + (-0.25)^2}{6-1}}$$

$$= 0.399\ A$$

v) Variance, $V = S^2 = (0.399)^2 = 0.115\ A^2$.

2.2.3 Probability of Errors

a) Normal distribution of errors

The normal or Gaussian law of errors is the basis for the major part of study of random effects. This type of distribution is most frequently met in practice. The law of probability states the normal occurrence of deviations from average value of an infinite number of measurements or observations can be expressed by,

$$y = \frac{h}{\sqrt{\pi}} \cdot e^{-h^2x^2} \qquad (1)$$

where, x = Magnitude of deviation

h = A constant called precision index

y = Number of readings at any deviation x, (the probability of occurrence of deviation x).

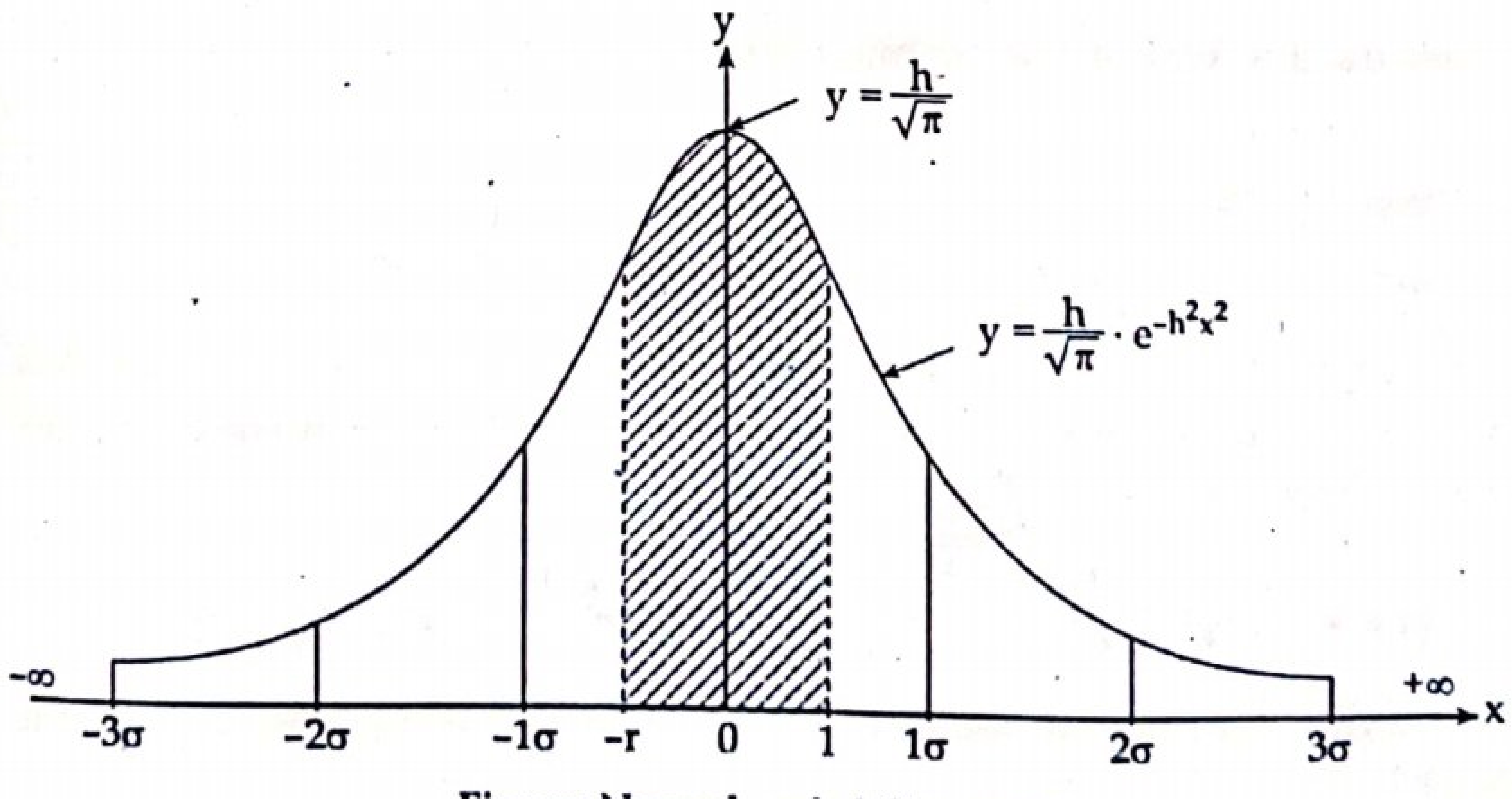


Figure: Normal probability curve

This curve is symmetrical about the arithmetic mean value, and area under the curve is unity. Under the conditions specified here the total number of readings taken is represented by 1.

From equation (1); when x = 0, $y = \frac{h}{\sqrt{\pi}}$

Thus it is clear from above that the maximum value of y depends upon h. The larger the value of h, the sharper the curve. Thus the value of h determines the sharpness of the curve since the curve drops sharply, owing to the term $(-h^2)$ being in the exponent. A sharp curve evidently indicates that the deviations are more closely grouped together around deviation x = 0.

b) Probable error

We know that the most probable or best value of a Gaussian distribution is obtained by taking arithmetic mean of the various values of the variate. In addition, it has been indicated that the confidence in this best value (most probable value) is connected with the sharpness of the distribution curve. A convenient measure of precision is the quantity r. It is called probable error or simple. P.E. The location of point r can be found from equation.

$$n_{1-2} = \frac{h}{\sqrt{\pi}} \int_{x_1}^{x_2} e^{-h^2x^2}\, dx$$

By putting $\frac{h}{\sqrt{\pi}} \int_{-r}^{+r} e^{-h^2x^2}\, dx = \frac{1}{2}$

This gives

$$r = \frac{0.4769}{h}$$

The standard deviation is given by, $\sigma^2 = \frac{\Sigma d^2}{n}$

The standard deviation for normal curve is $\sigma = \frac{1}{\sqrt{2h}}$

$$\sigma = \frac{r}{0.6745}$$

Thus, PE = r = ± 0.6745 σ

For an infinite number of deviations forming the normal probability curve, where n is infinite. But for a finite number of deviations, the probable error for one reading is:

$$r_1 = \pm 0.6745 \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + \dots\dots\dots + d_n^2}{n-1}} = \pm 0.6745 \sqrt{\frac{\Sigma|d|^2}{n-1}}$$

With a finite number of readings, the average reading has a probable error of

$$r_m = \frac{1}{\sqrt{n}} r_1 = \pm 0.6745 \sqrt{\frac{\Sigma|d^2|^2}{n(n-1)}}$$

The above equations means that for n finite readings, the probable error is r_m. If we have $n > 1$, then $n - 1 \simeq 1$ and $r_1 \simeq \pm 0.6745\ \sigma$ and $r_m = \pm 0.6745 \times \frac{\sigma}{\sqrt{n}}$.

Example 2

The following 10 observations were recorded when measuring a voltage 41.7, 42.0, 41.8, 42.0, 42.1, 41.9, 42.0, 41.9, 42.5 and 41.8 volt. Find the,

i) Mean
ii) Standard deviation
iii) Range
iv) Probable error of one reading
v) Variance

Solution:

i) Mean $(\overline{x}) = \frac{\Sigma x}{n}$

$$= \frac{41.7 + 42 + 41.8 + 42 + 42.1 + 41.9 + 42 + 41.9 + 42.5 + 41.8}{10}$$

$= 41.97$ V

ii) Standard deviation, $S = \sqrt{\frac{\Sigma d_t^2}{n-1}}$ for $n < 20$

Measured voltage (x)	Deviation (d)	d^2
41.7	– 0.27	0.0729
42.0	0.03	0.0009
41.8	0.17	0.0289
42.0	0.03	0.0009
42.1	0.13	0.0169
41.9	– 0.07	0.0049
42.0	0.03	0.0009
41.9	– 0.07	0.0049
42.5	0.53	0.2809
41.8	– 0.17	0.0289
$\Sigma x = 419.7$		$\Sigma d^2 = 0.441$

Thus, $S = \sqrt{\frac{0.441}{10-1}} = 0.22$ volt.

iii) Probable error of one reading,

$$r_1 = \pm 0.6745 \times S = \pm 0.6745 \times 0.22 = \pm 0.15 \text{ V}$$

iv) Probable error of mean,

$$r_m = \frac{\pm r_1}{\sqrt{n-1}} = \frac{\pm 0.15}{\sqrt{10-1}} = \pm 0.05 \text{ V}$$

v) Range = (Largest – Smallest) value = 42.5 – 41.7 = 0.8 volt

vi) Variance, $V = S^2 = (0.22)^2 = 0.0484$ volt^2

c) Limiting errors or Guarantee errors

The accuracy and presion of an instrument depends upon its design, the materials used and the workmanship that goes into making the instruments. The choice of an instrument for a particular application depends upon the accuracy desired. If only a fair degree of accuracy is desired, it is not economical to use expensive materials and skill into the manufacture of the instrument. But an instrument used for an application required a high degree of accuracy has to use expensive material and highly skilled workmanship.

The economical production of any instrument requires the proper choice of material, design and skill. Thus, the manufacturer has to specify the deviations from the nominal value of a particular quantity. The limits of these deviations from the specified value are defined as limiting errors.

The magnitude of a quantity having a nominal value as and a maximum error or limiting error of $\pm \delta A$ must have a magnitude Aa between the limits $A_s - \delta A$ and $A_s + \delta A$ or actual value, $A_a = A_s \pm \delta A$.

For example; the nominal magnitude of a resistor is 100 Ω with a limiting error of ± 10 Ω. The magnitude of the resistor will be between the limits:

$A = 100 \pm 10\ \Omega$ or $A > 90$ and $A < 110\ \Omega$. In other words, the manufacturer guarantees that the value of resistance of the resister lies between 90 Ω and 110 Ω.

Relative limiting error, $\epsilon_r = \frac{\delta A}{A_s} = \frac{\epsilon_0}{A_s}$

or, $\epsilon_0 = \delta A = \epsilon_r A_s$

Then,

$$A_a = A_s \pm \delta A = A_s \pm \epsilon_r A_s = A_s (1 + \epsilon_r)$$

∴ Percentage limiting error, $\% \epsilon_r = \epsilon_r \times 100$

∴ Relative limiting error,

$$\epsilon_r = \frac{A_a - A_s}{A_s} = \frac{\text{Actual value} - \text{Nominal value}}{\text{Nominal value}}$$

Example 3

A 0 – 150 V voltmeter has a guaranteed accuracy of 1% full scale reading. The voltage measured by this instrument is 83 V. calculate the limiting error in percentage.

Solution:

Magnitude of the limiting error is 1% of 150 V = $\frac{150}{100}$ = 1.5 V

% error of a meter indication of 83 V is $\frac{1.5}{83} \times 100\% = 1.81\%$

Example 4

Three resistors have the following ratings,

$R_1 = 37\ \Omega \pm 5\%$

$R_2 = 75\ \Omega \pm 5\%$

$R_3 = 50\ \Omega \pm 5\%$

Determine the magnitude and limiting error in ohm and in percent of the resistance of these resistances connected in series.

Solution:

The values of resistances are;

$$R_1 = 37 \pm \frac{5}{100} \times 37 = 37 \pm 1.85\ \Omega$$

$$R_2 = 75 \pm \frac{5}{100} \times 75 = 75 \pm 3.75\ \Omega$$

$$R_3 = 50 \pm \frac{5}{100} \times = 50 \pm 2.5\ \Omega$$

The limiting value of resultant resistance is,

$$R = (37 + 75 + 50) \pm (1.85 + 3.75 + 2.5) = 162 \pm 8.1\ \Omega$$

∴ Magnitude of resistance = $162\ \Omega$

Error is ohm = $\pm 8.1\ \Omega$

Percent limiting error = $\pm \frac{8.1}{162} \times 100 = \pm 5\%$

Example 5

The resistance of an unknown resistor is determined by the wheat stone bridge method. The solution for the unknown resistance is stated as

$$R_x = \frac{R_1 \times R_2}{R_3}$$

where, $R_1 = 500\ \Omega \pm 1\%$

$R_2 = 615\ \Omega \pm 1\%$

$R_3 = 100\ \Omega \pm 0.5\%$

Calculate:

i) Nominal value of the unknown resistor

ii) Limiting error in Ω of the unknown resistor

iii) Limiting error in percentage of the unknown resistor

Solution:

$R_1 = 500 \pm 5\ \Omega$

$R_2 = 615 \pm 6.15\ \Omega$

$R_3 = 100 \pm 0.5\ \Omega$

$$R_x = \frac{R_1 \times R_2}{R_3}$$

i) Nominal value of unknown resistor,

$$R_x = \frac{R_1 \times R_2}{R_3} = \frac{500 \times 615}{100} = 3{,}075\ \Omega$$

ii) Limiting error in ohms of the R_x, the value of R_1 and R_2 must be high and the value of R_3 should be low.

$$i.e.,\quad R_x = \frac{(500 + 5) \times (615 + 6.15)}{(100 - 0.5)} = \frac{505 \times 621.5}{99.5} = 3{,}152.57\ \Omega.$$

Now the limiting error is nominal value minus highest value of unknown resistor *i.e.,*

Limiting error $= (3{,}075 - 3{,}152.27)\ \Omega = -77.57\ \Omega$

Hence, the limiting error of unknown resistor is $\pm$ 77.57 Ω

iii) % limiting error of unknown resistor

$$= \frac{\text{Limiting error}}{\text{Nominal value}} \times 100\% = \frac{77.57}{3{,}075} \times 100\% = \pm 2.52\%$$

2.3 PERFORMANCE PARAMETERS (STATIC AND DYNAMIC)

The treatment of instrument and measurement system characteristics can be divided into two distinct categories viz;

a) Static characteristics

b) Dynamic characteristics

2.3.1 Static Characteristics

The static characteristics of a measurement system are those that must be considered when the system or instrument is used to measure a conditions not varying with time *i.e.,* when steady state condition occurs some of its static performance parameter are as follows;

i) Accuracy

The accuracy of an instrument is a measure of how close the output readings of the instrument is to the correct value. Thus accuracy of a measurement means conformity to truth. In practice, it is more usual to quote the inaccuracy figure rather than the accuracy figure for an instrument. Inaccuracy is the extent to which a reading might be wrong, and is often quoted as a percentage of the full scale reading of an instrument.

ii) Precision/repeatability/reproducibility

Precision is a term that describe an instrument's degree of freedom from random errors. If a large number of readings are taken of the same quantity by a high precision instrument, then the spread of readings will be very small. High precision does not imply anything about measurement accuracy. A high precision instrument may have a low accuracy.

Repeatability describes the closeness of output, readings when the same input is applied repetitively over a short period of time, with the same measurement conditions, same instrument and observer, same location and same conditions of use maintained throughout.

Reproducibility describes the closeness of output readings for the same input when there are changes in the method of measurement, observer, measuring instrument, location, conditions of use and time of measurement.

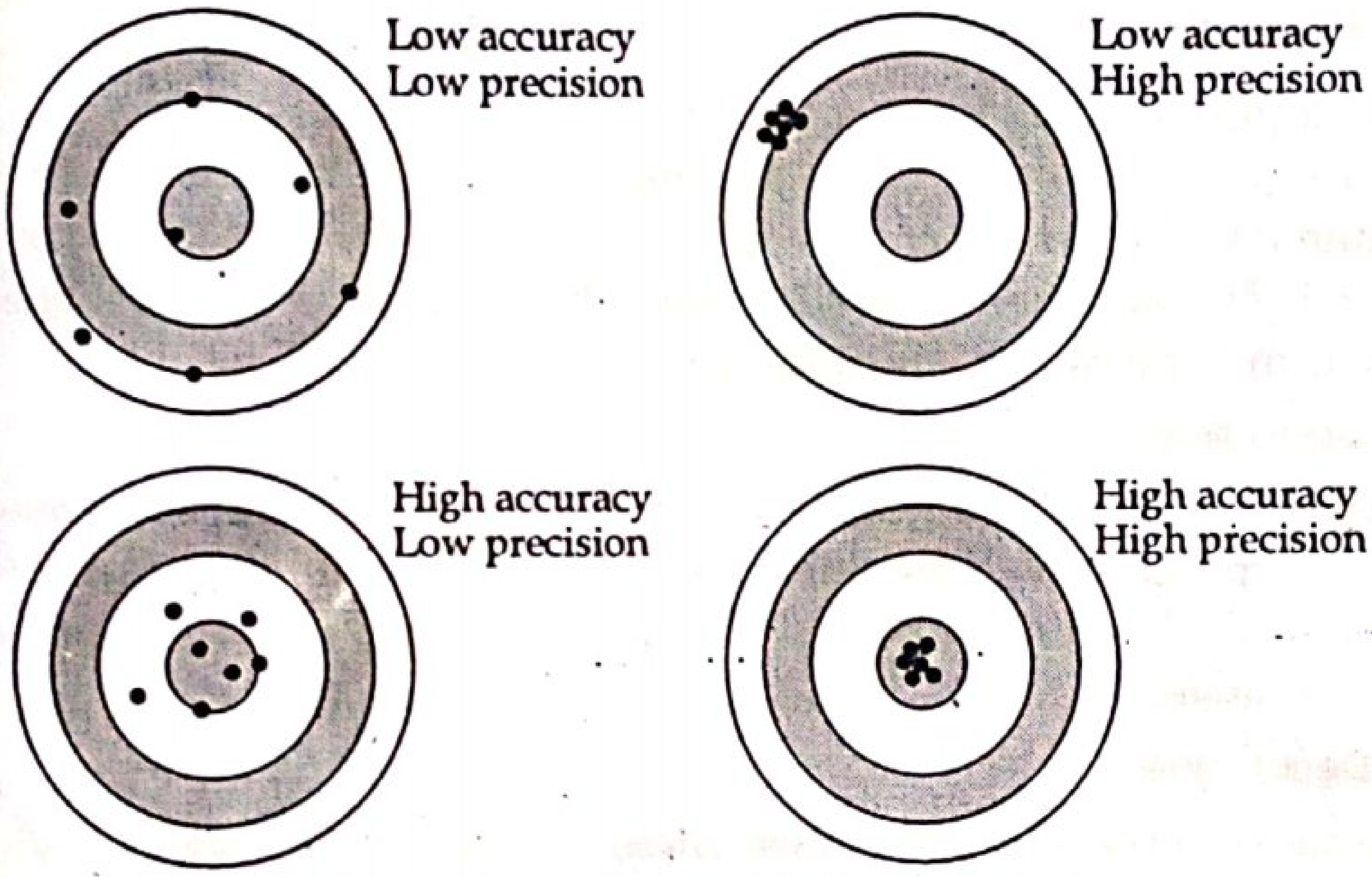


Figure: Accuracy and precision comparison

iii) Linearity

One of the best characteristics of an instrument or a measurement system is considered to be linearity, that is, the output is linearly proportional to the input. Most of the system require a linear behavior as it is desirable. This is because the conversion from a scale reading to the corresponding measured value of input quantity is most convenient if one merely has to multiply by a fixed constant rather than consult a non-linear calibration curve or compute from non-linear calibration equations.

It is represented as $y = mx + c$

where, y = output

x = input

m = slope

c = intercept

iv) Hysteresis

Hysteresis effects shows up in any physical, chemical or electrical phenomenon. Hysteresis is a phenomenon which depicts different output effects when loading and unloading whether it is a mechanical system or an electrical system and for that matter any system. Hysteresis is non-incidence of loading and unloading curves. Hysteresis is most commonly found in instruments that contain springs such as the passive pressure gauge.

Hysteresis, in a system, arises due to the fact that all the energy put into the stressed parts when loading is not recoverable upon loading. This is because second law of thermodynamics rules out any perfectly reversible process in the world.

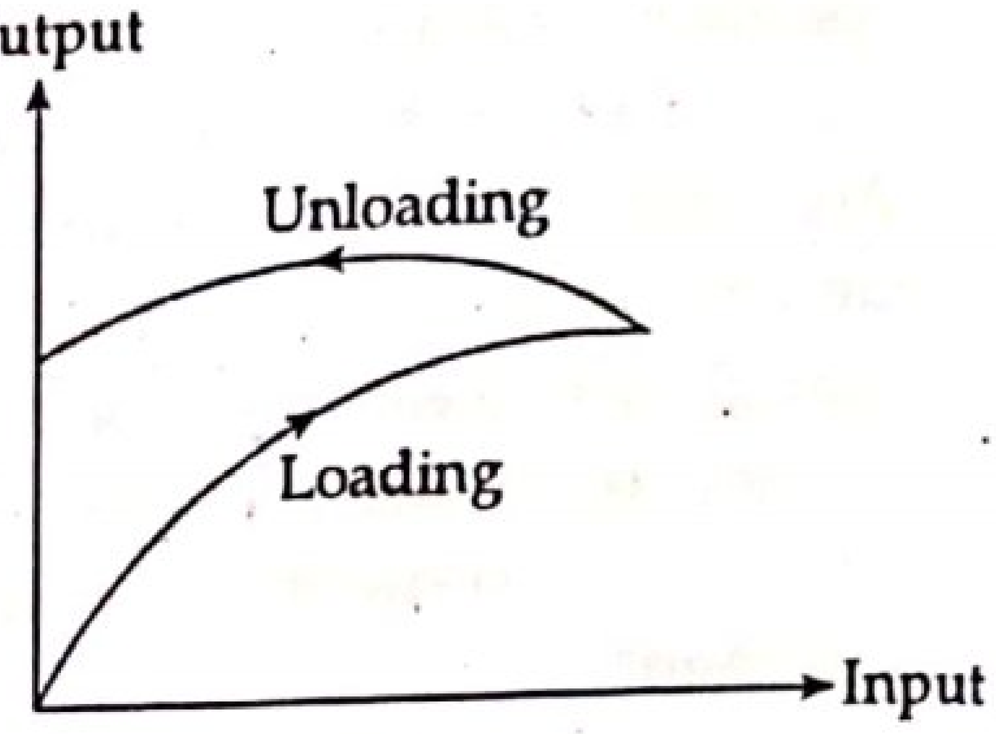


Figure: Hysteresis curve

v) Threshold

If the input to an instrument is gradually increased from zero, the input will have to reach a certain minimum level before the change in the instrument output reading is of a large enough magnitude to be detectable. This minimum level of input is known as the threshold of the instrument.

vi) Dead time

Dead time is defined as the time required by a measurement system to begin to respond to a change in a measurand. Dead time, in fact, is the time before the instrument begins to respond after the measured quantity has been changed.

vii) Dead zone

Dead zone is defined as the largest change of input. Quantity for which there is no output of the instrument. Any instruments that exhibits hysteresis also displays dead space.

viii) Resolution or discrimination

The smallest increment in input (the quantity being measured) which can be detected with certainty by an instrument is its resolution or discrimination. So resolution defines the smallest measurable input changes while the threshold defines the smallest measurable input.

ix) Loading effect

The incapatability of the system to faithfully measure, record or control the input signal (measured) in undistorted form is called the loading effect.

x) Range or span

The range or span of an instrument defines the minimum and maximum values of a quantity that the instrument is designed to measure.

xi) Sensitivity

The sensitivity of measurement is a measure of the change in instrument output that occurs when the quantity being measured changes by a given